LAPORAN WORKSHOP DEVELOPER OPERATIONAL

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Golongan B

PROGRAM STUDI TEKNIK INFORMATIKA

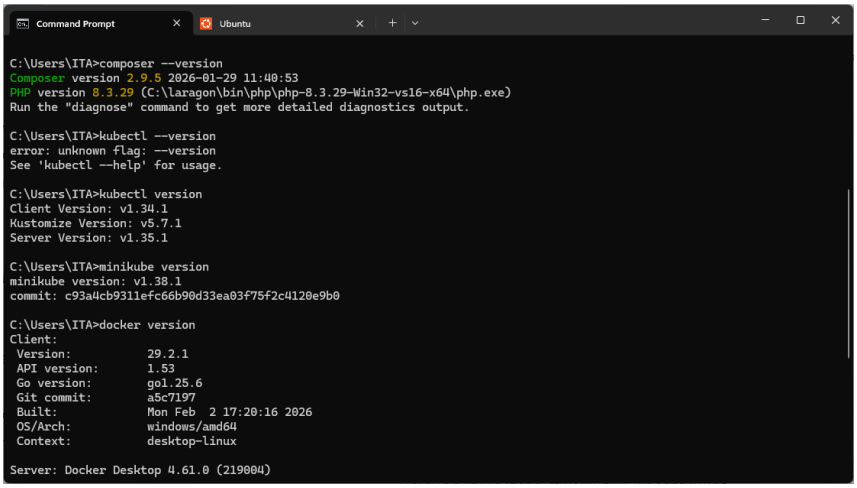
JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI JEMBER

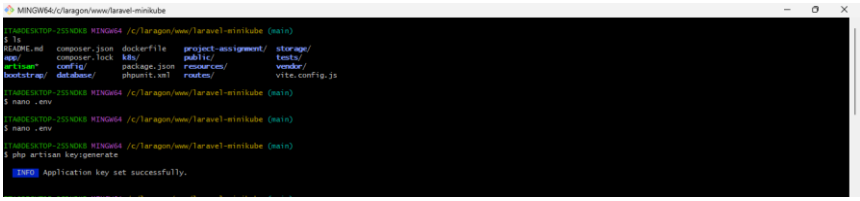
2026

1.1 Hasil Praktikum Deployment aplikasi Laravel menggunakan Docker dan Kubernetes

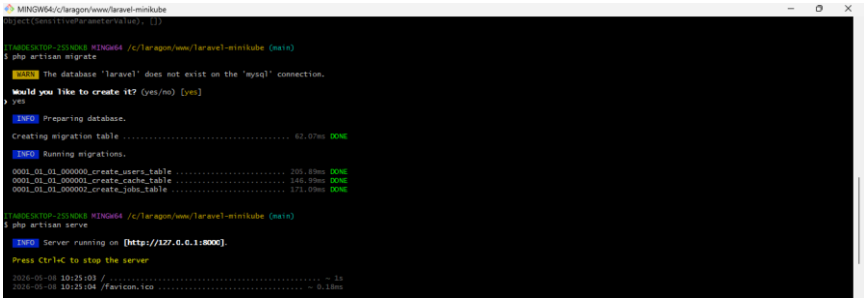
- Memastikan PHP 8.3 atau versi terbaru, Composer, Docker Desktop, Minikube, Kubectl dan laragon ter install



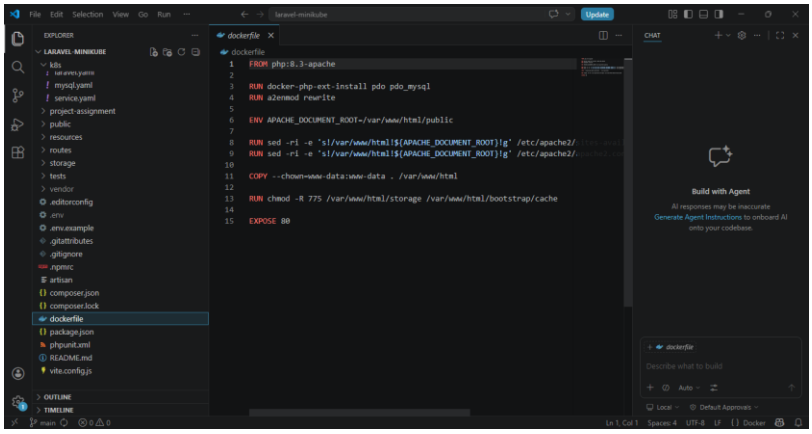
- Melakukan set up proyek Laravel



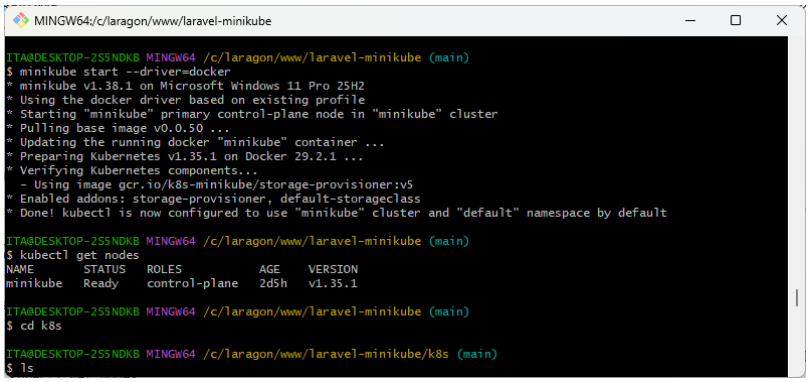
- Melakukan pengujian di lokal



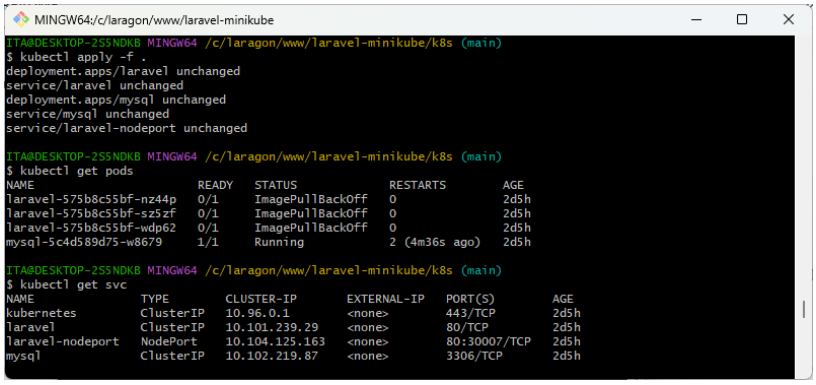
- Konfigurasi docker file



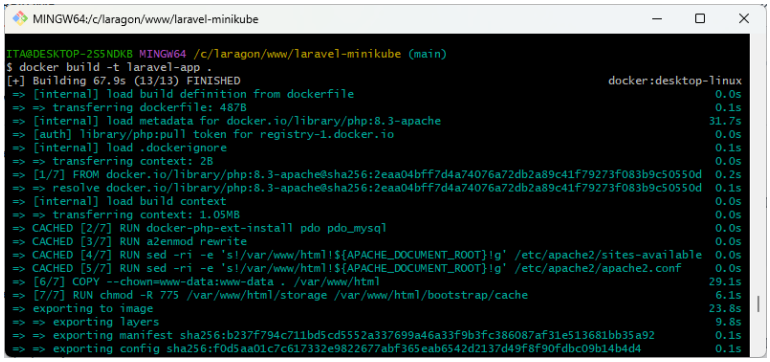
- Inisialisasi Kubernetes (Minikube)

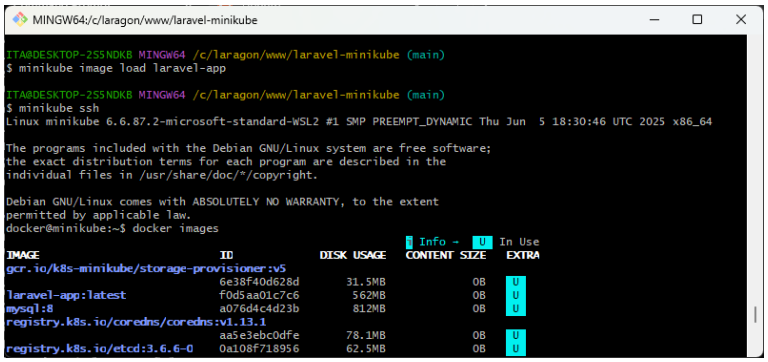


- Setup Kubernetes Manifest

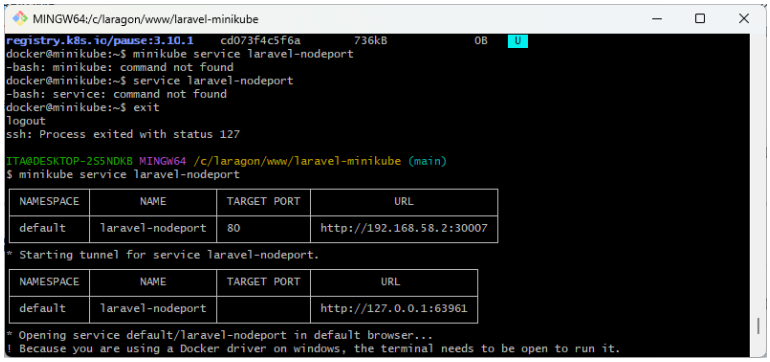


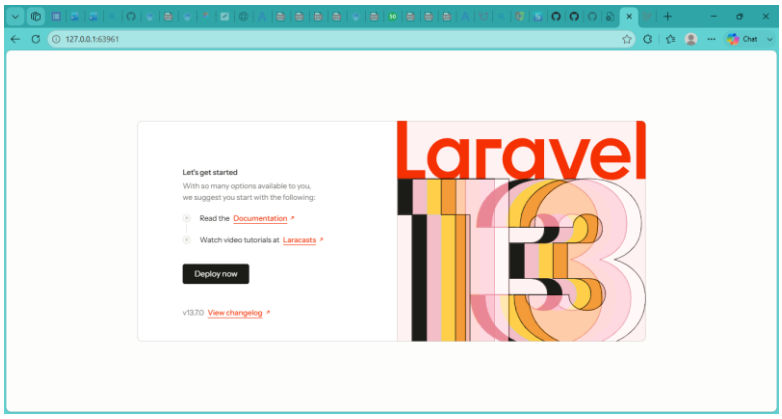
- Build dan Load Image ke Minikube





- Menjalankan Aplikasi





- Migrasi Database di Pod

```
MINGW64:/c/laragon/www/laravel-minikube

ITA@DESKTOP-2S5NDKB MINGW64 /c/laragon/www/laravel-minikube (main)
$ kubectl get pods
NAME                        READY   STATUS    RESTARTS      AGE
laravel-575b8c55bf-nz44p    1/1     Running   0             2d5h
laravel-575b8c55bf-sz5zf    1/1     Running   0             2d5h
laravel-575b8c55bf-wdp62    1/1     Running   0             2d5h
mysql-5c4d589d75-w8679      1/1     Running   2 (12m ago)   2d5h

ITA@DESKTOP-2S5NDKB MINGW64 /c/laragon/www/laravel-minikube (main)
$ kubectl exec -it <laravel-^C-- bash

ITA@DESKTOP-2S5NDKB MINGW64 /c/laragon/www/laravel-minikube (main)
$ kubectl exec -it <laravel-575b8c55bf-nz44p> -- bash
bash: laravel-575b8c55bf-nz44p: No such file or directory

ITA@DESKTOP-2S5NDKB MINGW64 /c/laragon/www/laravel-minikube (main)
$ kubectl exec -it laravel-575b8c55bf-nz44p -- bash
root@laravel-575b8c55bf-nz44p:/var/www/html# php artisan migrate

  INFO  Preparing database.

  Creating migration table ..................................... 371.41ms DONE
```

- Manajemen Skalabilitas

```
MINGW64:/c/laragon/www/laravel-minikube

ITA@DESKTOP-2S5NDKB MINGW64 /c/laragon/www/laravel-minikube (main)
$ kubectl get pods -w
NAME                        READY   STATUS    RESTARTS      AGE
laravel-575b8c55bf-nz44p    1/1     Running   0             2d5h
laravel-575b8c55bf-sz5zf    1/1     Running   0             2d5h
laravel-575b8c55bf-wdp62    1/1     Running   0             2d5h
mysql-5c4d589d75-w8679      1/1     Running   2 (17m ago)   2d5h


ITA@DESKTOP-2S5NDKB MINGW64 /c/laragon/www/laravel-minikube (main)
$ kubectl scale deployment laravel --replicas=5
deployment.apps/laravel scaled

ITA@DESKTOP-2S5NDKB MINGW64 /c/laragon/www/laravel-minikube (main)
$ kubectl get pods -w
NAME                        READY   STATUS    RESTARTS      AGE
laravel-575b8c55bf-4zskf    1/1     Running   0             6s
laravel-575b8c55bf-kr4jn    1/1     Running   0             6s
laravel-575b8c55bf-nz44p    1/1     Running   0             2d5h
laravel-575b8c55bf-sz5zf    1/1     Running   0             2d5h
laravel-575b8c55bf-wdp62    1/1     Running   0             2d5h
mysql-5c4d589d75-w8679      1/1     Running   2 (18m ago)   2d5h
```

```
MINGW64:/c/laragon/www/laravel-minikube


ITA@DESKTOP-2S5NDKB MINGW64 /c/laragon/www/laravel-minikube (main)
$ kubectl scale deployment laravel --replicas=2
deployment.apps/laravel scaled

ITA@DESKTOP-2S5NDKB MINGW64 /c/laragon/www/laravel-minikube (main)
$ kubectl get pods -w
NAME                        READY   STATUS        RESTARTS      AGE
laravel-575b8c55bf-4zskf    1/1     Terminating   0             5s
laravel-575b8c55bf-kr4jn    1/1     Terminating   0             5s
laravel-575b8c55bf-nz44p    1/1     Running       0             2d5h
laravel-575b8c55bf-wdp62    1/1     Running       0             2d5h
mysql-5c4d589d75-w8679      1/1     Running       2 (18m ago)   2d5h
laravel-575b8c55bf-4zskf    0/1     Completed     0             41s
laravel-575b8c55bf-kr4jn    0/1     Completed     0             41s
laravel-575b8c55bf-4zskf    0/1     Completed     0             42s
laravel-575b8c55bf-4zskf    0/1     Completed     0             42s
laravel-575b8c55bf-kr4jn    0/1     Completed     0             42s
laravel-575b8c55bf-kr4jn    0/1     Completed     0             42s


ITA@DESKTOP-2S5NDKB MINGW64 /c/laragon/www/laravel-minikube (main)
$
```